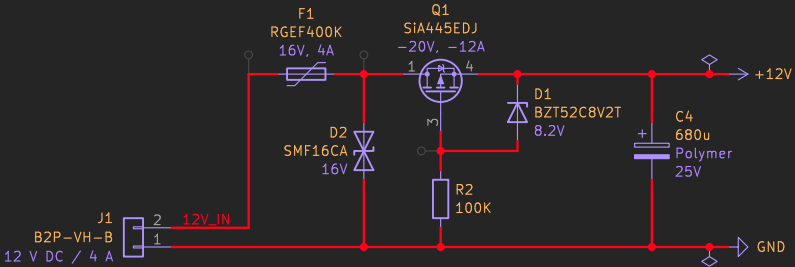


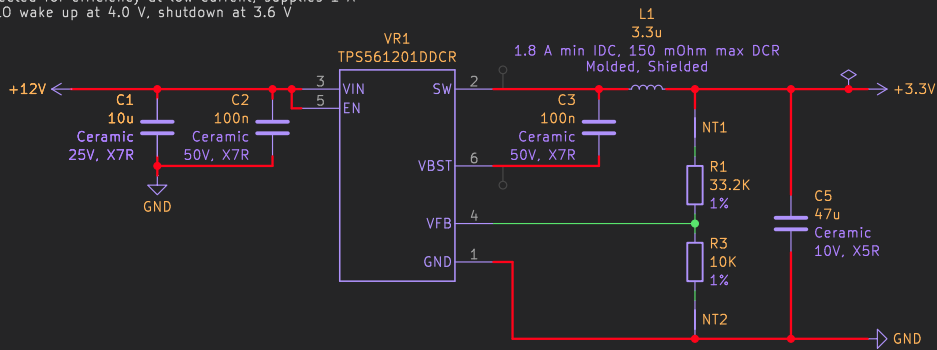
POWER INPUT

Input must be 9 to 16 V
Circuit protection
– Internal: 4 A polyfuse, 16 V TVS, reverse polarity
– External: 5 A fuse recommended

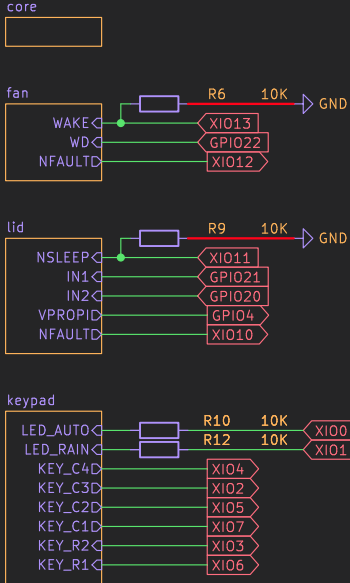


3.3 V BUCK CONVERTER

Selected for efficiency at low current, supplies 1 A
UVLO wake up at 4.0 V, shutdown at 3.6 V

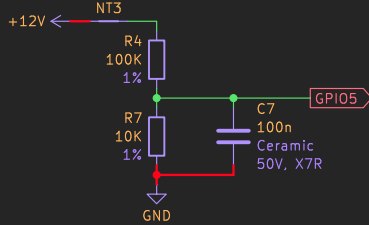


COMPONENTS



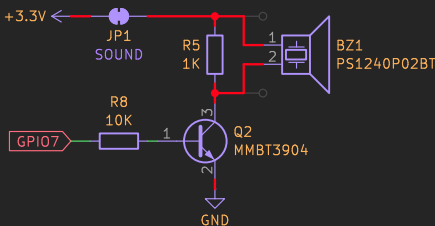
VOLTAGE SENSOR

Divide input voltage by 11 for ADC



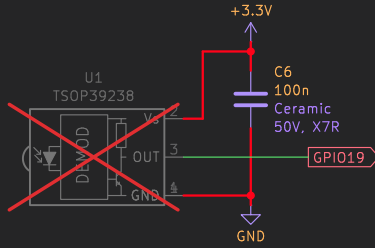
PIEZO BUZZER

Driven at 3.3 V for politeness (12 V would be louder)
Can be disabled in software or by cutting the solder jumper



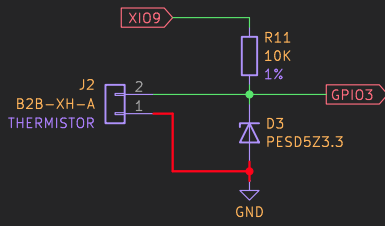
IR REMOTE RECEIVER

Operates on 38 kHz carrier frequency



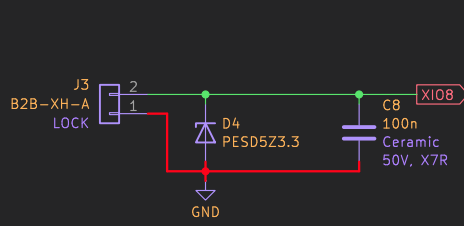
TEMPERATURE SENSOR

Built-in 10 K NTC thermistor, beta = 3950
Powered only while sampling to minimize self-heating



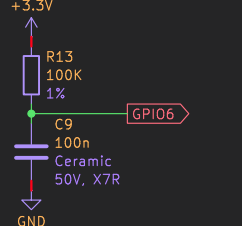
SAFETY LOCK

While the input is tied to ground, stops the fan,
closes the lid, and inhibits operation



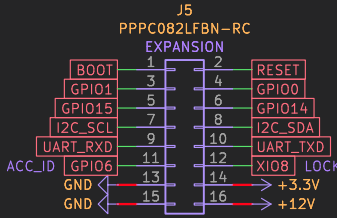
ACCESSORY ID

Identifies the expansion port accessory
Refer to the design document for details



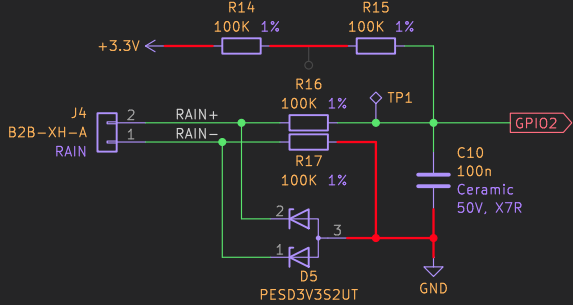
EXPANSION PORT

Header for accessories and factory programming
– GPIO0, GPIO1, GPIO14, GPIO15 can be used for any purpose
– UART_RXD and UART_TXD provide the serial port
– I2C_SCL and I2C_SDA provide the I2C bus (QWIIC)
– RESET and BOOT are wired in parallel with their corresponding buttons (active low)
– LOCK engages the safety lock (active low)
– ACC_ID identifies the expansion port accessory
– 12 V supply is unregulated, 1 A current available
– 3.3 V supply is regulated, 600 mA current available

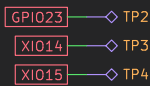


RAIN SENSOR

The rain sensor is only available on model 4500K; these components may be omitted for other models
It is simply a pair of electrodes that are exposed to the environment and whose impedance is about 0.5 to 1.2 Mohm when in contact with water or mist and open circuit when dry
With these assumptions, the ADC should read less than ~2.9 V when wet



PINS RESERVED FOR FUTURE USE



github.com/brown-studios/minuet

Brown Studios LLC

Sheet: /

File: minuet.kicad_sch

Title: Minuet Fan Controller

Size: A3

Date: 2026-04

Rev: v4.0

KiCad E.D.A. 9.0.6

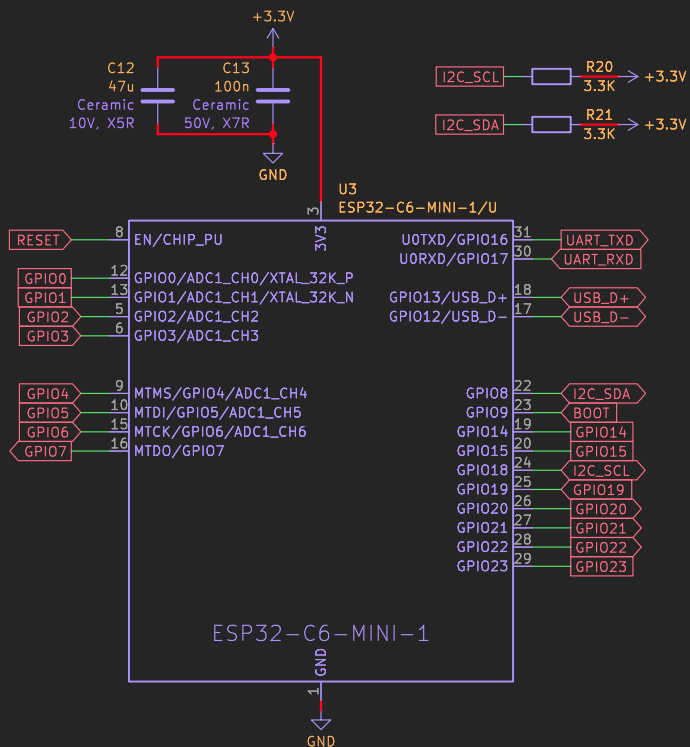
Id: 1/5

MICROCONTROLLER

ESP32-C6 strapping pins

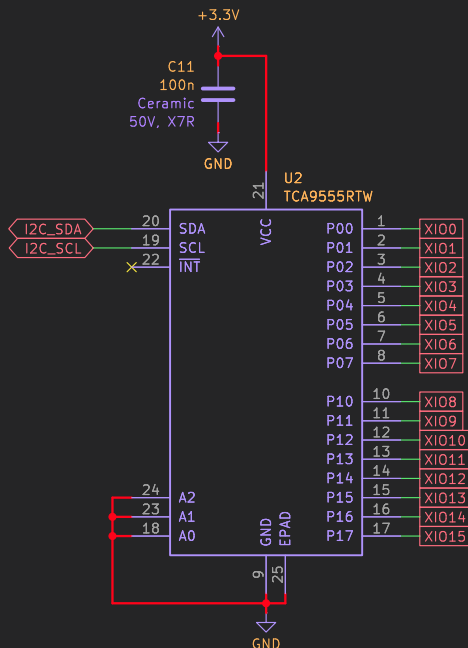
- GPIO4 and GPIO5: don't care because SDIO is unused
- GPIO8: must be high for download mode, re-used for I2C
- GPIO9: boot select, high to boot from flash, low for download mode
- GPIO15: don't care because EFUSE_JTAG_SEL_ENABLE = 0

Internal pull-up active on reset: GPIO6, GPIO9, GPIO13, GPIO16-23



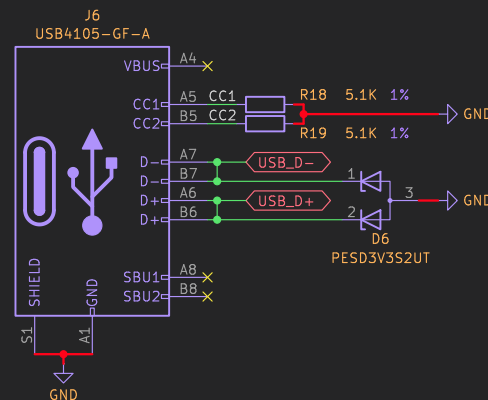
IO EXPANDER

XIO pins have internal 100 K pull-ups, configured as inputs at reset, cannot be set to Hi-Z, 50 mA push-pull output, 5 V tolerant



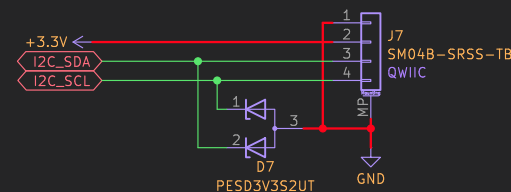
USB-C INTERFACE

ESP32-C6 only supports full-speed data rate (11 Mbps)



QWIC INTERFACE

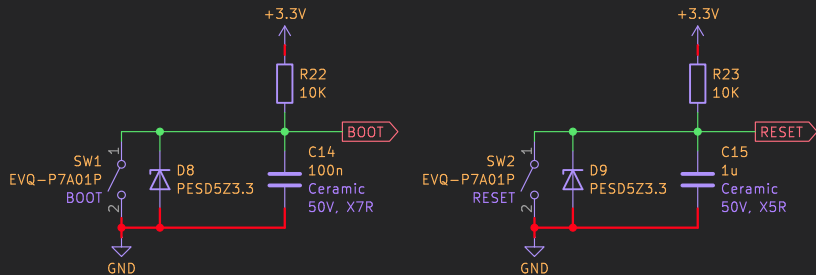
A great place to plug in off-the-shelf accessories
- Host has pull-ups so accessories don't need their own
- 3.3 V supply is regulated, 600 mA current available



BUTTONS

Hold BOOT, press/release RESET, then release BOOT to enter the bootloader for firmware updates

RESET forms an RC delay circuit with R = 10 Kohm, C = 1 uF to ensure power supply stability before boot



github.com/brown-studios/minuet

Brown Studios LLC

Sheet: /core/

File: core.kicad_sch

Title: Minuet Core Logic

Size: A4

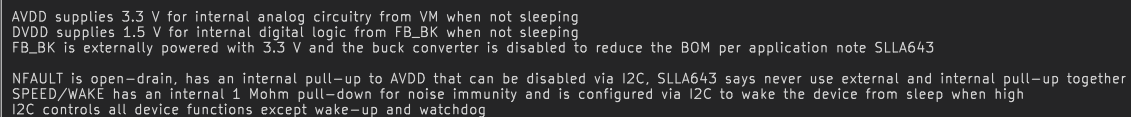
Date: 2026-04

KiCad E.D.A. 9.0.6

Rev: v4.0

Id: 2/5

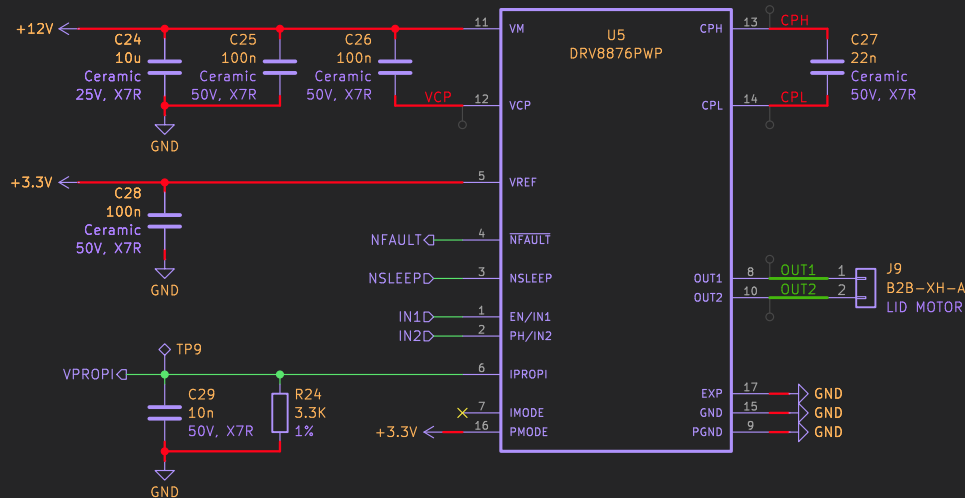
Designed for 12 V BLDC motor at up to 4 A



Id: 3/5

LID MOTOR DRIVER

Designed for 12 V DC motor at up to 2 A



PMODE set to PWM mode with IN1 / IN2 signals
 IMODE set to mode 4: fixed off time current chopping, overcurrent latches output off and reports a fault
 $I_{trip} (A) = 1000 * V_{ref} (V) / R_{propi} (\Omega) = 1 A$
 NSLEEP activates motor when high
 NFAULT is open-drain

Internal pull-down on NSLEEP, IN1, IN2

github.com/brown-studios/minuet

Brown Studios LLC

Sheet: /lid/

File: lid.kicad_sch

Title: Minuet Lid Motor Driver

Size: A4

Date: 2026-04

KiCad E.D.A. 9.0.6

Rev: v4.0

Id: 4/5

KEYPAD AND WIRED WALL CONTROL INTERFACE

Compatible with original equipment

Key matrix

- R1+GND: Close lid
- R2+GND: Open lid
- R1+C1: Fan off
- R1+C2: Fan speed up
- R1+C3: Fan speed down
- R1+C2+C3: Open / close lid
- R1+C4: Toggle rain sensor (only certain models)
- R2+C1: Fan on and cycle speeds
- R2+C2: Fan on / off
- R2+C3: Fan intake / exhaust direction
- R2+C4: Toggle auto thermostat

LEDs

- LED_AUTO: Auto LED cathode, green with 2.4 V voltage drop
- LED_RAIN: Rain LED cathode, red with 2 V voltage drop (only certain models)

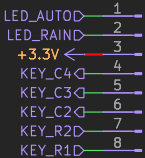
KEY_Rx and KEY_Cx inputs need pull-ups

Detect key press when KEY_Rx goes low when undriven (tristate)

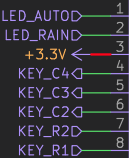
Detect key press when KEY_Cx goes low during high-to-low strobe of KEY_Rx

Fan originally provided +5V supply the interface but +3.3V works fine

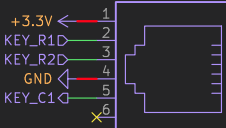
J10
G800LE300018YEU



J11
MTJ-889X1



J12
MTJ-662BX2



A note on 8P8C (RJ45) and 6P6C (RJ12/RJ11) jacks:

These jacks are available in numerous footprints and form factors. For this application, we prefer a jack that is sealed on the top to keep out condensation and a tab on the bottom for ease of access such as MTJ-889X1 and MTJ-669X1. However, while these specific parts are readily available at Digikey, they are not normally stocked and relatively expensive at JLCPCB so we must also consider alternatives for manufacturing.

Selected equivalent parts:

8P8C (tab down, 17.3 mm OAL)

- Adam Tech MTJ-889X1
- Amphenol 54602-908LF
- HCTL HC-RJ45-058-1-6
- XunPu RJ45-306-005-18
- XKB X31ADIWA3DY1027

6P6C (tab up, 13 mm OAL):

- Adam Tech MTJ-662BX1/MTJ-662BX2
- CONNPLY Elec DS1133-S60BPX
- Ckmtw R-RJ11R06P-A800

github.com/brown-studios/minuet

Brown Studios LLC

Sheet: /keypad/

File: keypad.kicad_sch

Title: Minuet Keypad Interface

Size: A4

Date: 2026-04

KiCad E.D.A. 9.0.6

Rev: v4.0

Id: 5/5